# Harshit Mangal

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## EDUCATION

**MBM University** Aug. 2023 – May 2027
*Bachelor of Technology in Information Technology — **CGPA: 7.2*** Jodhpur, Rajasthan

## PROJECTS

**ShopSphere — AI-Driven B2C E-Commerce Platform [Live Demo | Frontend | Backend]** 2026
*React, Node.js, Express, MongoDB, Socket.io, Redis, Redux Toolkit, Google Gemini 2.5 Flash, Razorpay, PayPal*

- Built an AI-driven B2C e-commerce platform integrating a **Google Gemini** assistant (LangChain) for catalog queries and real-time translations, backed by resilient Redis caching with sub-2ms response latency.
- Architected real-time **Socket.io support chat** with room-based isolation and MongoDB aggregation pipelines for active threads.
- Integrated dynamic multi-currency payments (Razorpay, PayPal), group buying, and secure JWT / OAuth authentication sessions.

**AI Document Assistant (Offline RAG) [GitHub Repo]** 2025
*Python, Streamlit, Ollama (LLaMA 3.2 3B), FAISS, SentenceTransformers, EasyOCR, PyMuPDF*

- Architected a 100% offline, air-gapped RAG system using local **Ollama (LLaMA 3.2)** and SentenceTransformers, ensuring strict data privacy with zero external API dependencies.
- Engineered a hybrid document parser supporting multi-format files (PDF, Word, Images) with GPU-accelerated EasyOCR fallback.
- Implemented in-memory FAISS semantic vector search with 800-char chunking, enabling sub-second document retrieval on LAN.

**Wanderlust — Geospatial Rental Booking SaaS [Live Demo | GitHub Repo]** 2025
*React, Node.js, Express, MongoDB, Socket.io, Mapbox GL JS, Cloudinary*

- Developed a full-stack rental property marketplace with secure user authentication and property management workflows.
- Integrated **Mapbox GL JS & Geocoding APIs** for interactive geospatial property discovery, map search, and guest-to-host messaging.
- Engineered Cloudinary media upload pipelines with dynamic compression, reducing page payload by 60% and optimizing load speed.

## EXPERIENCE

**Frontend Web Development Intern** June 2025 – July 2025
*CodTech IT Solutions Pvt. Ltd.* Remote

- Independently engineered 3 production-grade applications: a real-time chat app, interactive quiz, and responsive e-learning UI.
- Implemented cross-browser compatible, mobile-first interfaces using HTML5, CSS3, JavaScript; integrated REST APIs and Git workflows.

## TECHNICAL SKILLS

**Languages:** Python, C++, C, Java, JavaScript, SQL

**Web & Frameworks:** HTML, CSS, React, Redux Toolkit, Node.js, Express, Streamlit, Tailwind CSS, Mongoose, JWT, Socket.io

**AI/ML & Databases:** RAG Architecture, Ollama (LLaMA 3.2), FAISS, Redis, MongoDB, MySQL, SQLite, SentenceTransformers

**Tools & CS Fundamentals:** Data Structures & Algorithms, OS & Concurrency, Docker (Basics), Git, GitHub, Postman, Linux, REST APIs

## ACHIEVEMENTS

- **LeetCode Knight:** Achieved prestigious **Knight Badge** with peak Contest Rating 1905–1931 (Top 3.8% globally, 1350+ solved).
- **GeeksforGeeks College Rank 1:** Secured **Rank 1** across MBM University with 1100+ problems solved.
- **Winner — Environmental Hacks:** 1st Place Winner at Environmental Hacks (WeMakeDevs & AWS) for cloud-native environmental tracking.
- **Amazon ML Challenge:** Secured **All India Rank 165** competing against nationwide engineering teams.

## CERTIFICATIONS

- 160 Days of Problem Solving — GeeksforGeeks (2025)
- Problem Solving (Intermediate) & Problem Solving (Basic) — HackerRank (2026)